Growth Model: Practice Session

Reps, reps, reps

PROFESSOR KIRSTEN CORNELSON · GLOBAL MACROECONOMICS FOR MANAGERS

IVEY

A provocative claim. A very provocative claim.

Opinion **Demographics and population**

Does Africa have too few people?

What some see as an alarming population explosion could be just what the continent needs



DAVID PILLING + Add to myFT



Populations are becoming concentrated enough to achieve the economic growth that occurs most readily in towns and cities, with Ethiopia racking up 15 years of 7 per cent growth © Amanuel Sileshi/AFP/Getty Images

David Pilling

Published AUG 11 2026

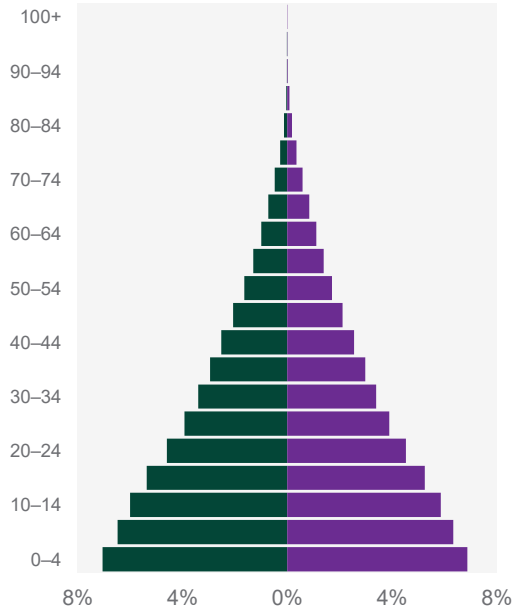
455

Africa is far younger: its working-age bulge is still to come

■ Men (left) ■ Women (right) Share of total population by five-year age group, 2025. Same scale in every panel.

Africa (whole continent)

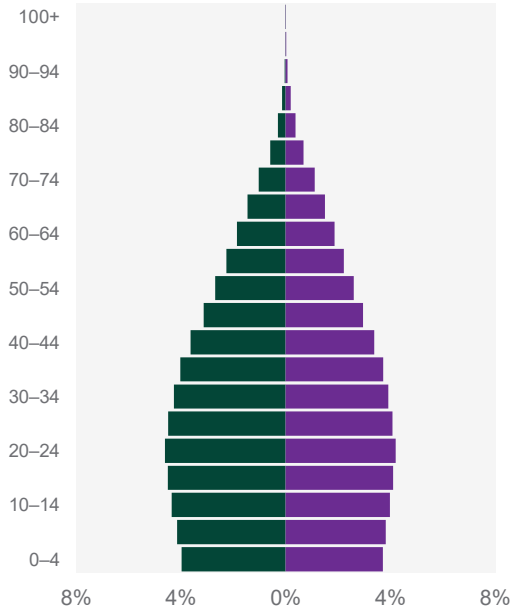
1.57 billion people



Median age	**20**
Under 15	**39%**
65 and over	**4%**
Dependents per 100 working-age	**73**

India

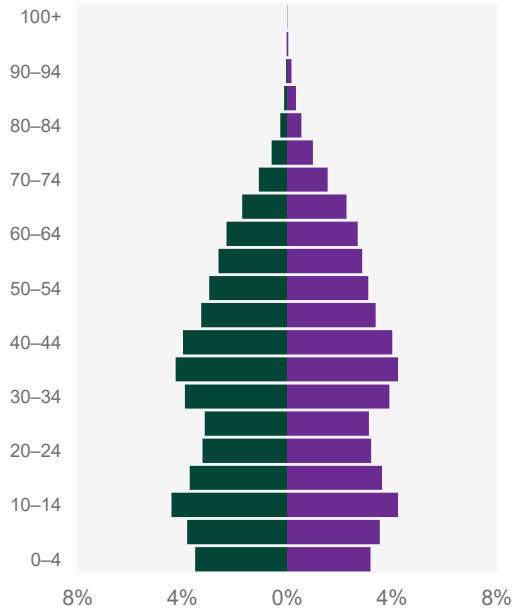
1.47 billion people



Median age	**30**
Under 15	**24%**
65 and over	**8%**
Dependents per 100 working-age	**46**

Vietnam

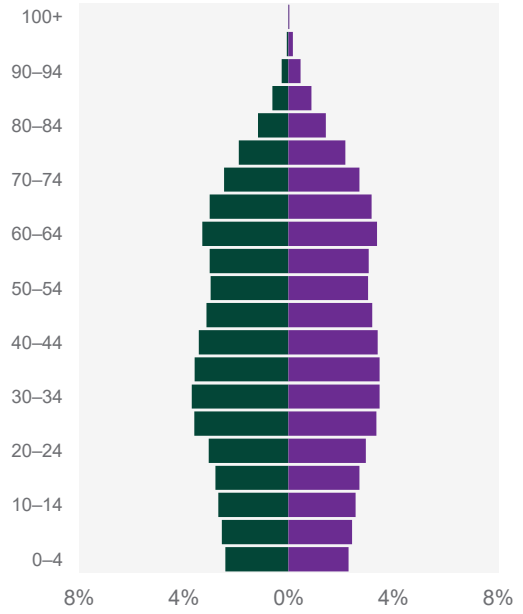
102 million people



Median age	**35**
Under 15	**23%**
65 and over	**10%**
Dependents per 100 working-age	**48**

Canada

40 million people



Median age	**42**
Under 15	**15%**
65 and over	**21%**
Dependents per 100 working-age	**55**

Source: United Nations, World Population Prospects 2024, medium-variant estimates for 2025 (wpp2024 data package). Our calculations.

WARM-UP · BUZZ · 5 MINUTES

FINANCIAL TIMES · OPINION

“Does Africa have too few people?”

The thesis: the most important reason for Africa’s relative poverty has been low population density, and fast population growth is now fixing it.

2.5bn

Africa’s projected population in 2050, up from 1.5bn

19

median age of Africa’s population

~4

children per woman, on average

How could this possibly be true? Use our growth model.

Source: Financial Times opinion column (2026) on Joe Studwell, How Africa Works: Success and Failure on the World’s Last Developmental Frontier (2026).

Too few people kept markets small and made roads, schools and states expensive

9×

Africa has about as many people as India on nine times the land

1500s

In 1975, Africa was as densely populated as Europe in the 1500s

2030

the year Africa's density reaches Asia's level of about 1960

1 **No agglomeration** — PRODUCTIVITY (TFP)

Scattered people meant small markets, thin labour pools and few cities: the places where specialization and learning happen.

2 **Costly infrastructure per person** — CAPITAL

Roads, power grids and water systems cost too much when each kilometre serves only a few people.

3 **Costly schools and clinics per person** — HUMAN CAPITAL

As with infrastructure, because people are thinly spread, too few children lived within reach of a school and too few families within reach of a clinic.

4 **Too sparse to govern or tax** — INSTITUTIONS

Scattered people were costly to reach, govern and tax, keeping states weak. Colonial rule focused on building extractive export enclaves around mines and plantations

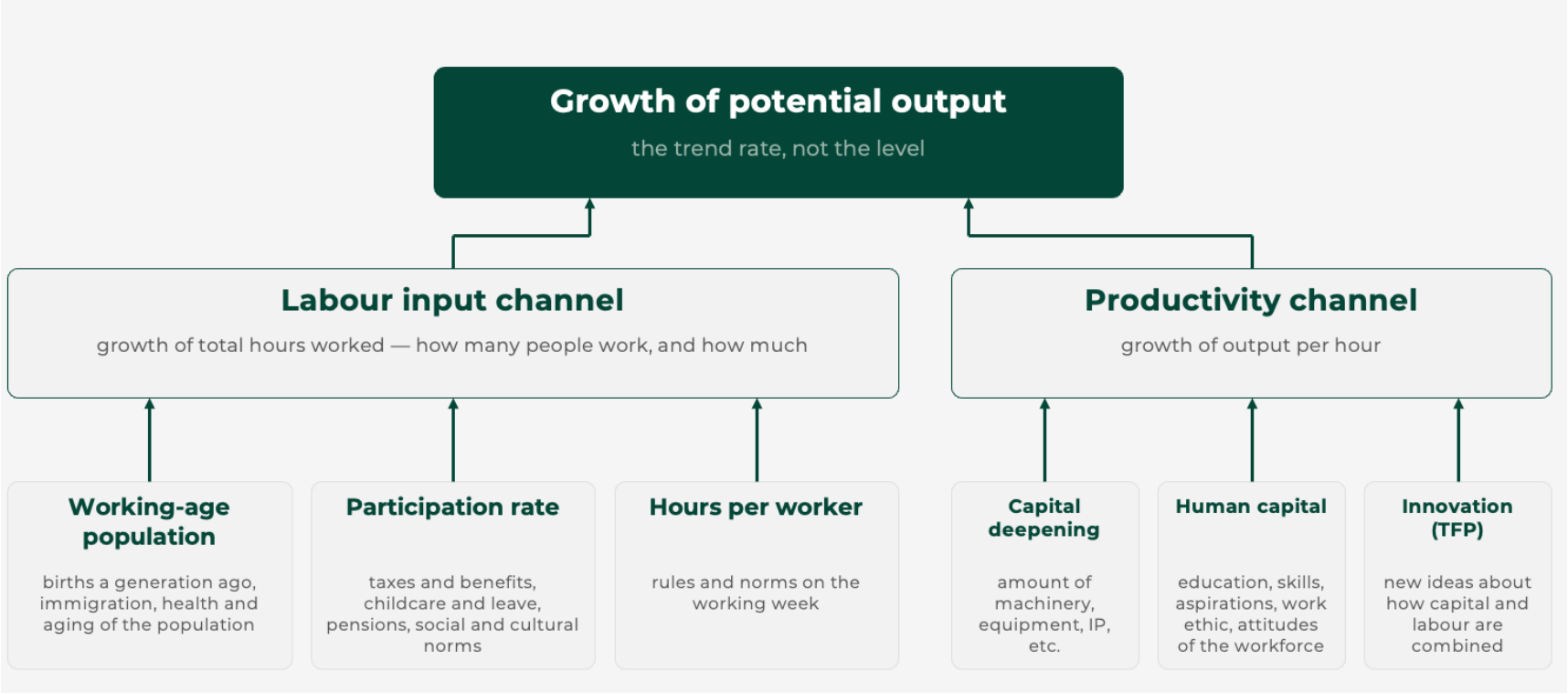
5 **Now catching up** — THE TURNING POINT

Disease, poor soils and the slave trade kept numbers low. Modern medicine lifted growth from 0.8% a year (1930s) to 2.9% (1980s).

Sources: Joe Studwell, How Africa Works (2026), as summarized in the Financial Times, The Economist and the Policy Center for the New South; population and density figures as reported there.

BUZZ · 5 MINUTES

Why might this argument be wrong?



Hint: Consider what a population growing close to 3% a year, with about four children per woman, does to the drivers of our growth model?

BUZZ · 5 MINUTES

Why might this argument be wrong?

1	**The evidence**	If density drives development, which countries should be rich, and are they?
2	**The direction**	Does density cause prosperity, or does prosperity cause density?
3	**The growth model**	Faster population growth adds people. But might it hinder productivity?

Density can help development, but it is neither necessary nor sufficient

1 The data don’t line up

EVIDENCE

Rwanda (566 people per km²) and Burundi (533) are among Africa’s densest countries and are low-income. Botswana and Namibia (4 per km²) are upper-middle-income. Sparse Canada and Australia are rich.

2 Causality may run the other way

INSTITUTIONS

Before industrialization, productive places supported more people. Among former colonies, those richest and densest in 1500 are relatively poor today: institutions, not density, made the difference.

3 More people is labour input, not productivity

LABOUR INPUT · CAPITAL

Rapid population growth spreads capital across more workers, and when labour is this cheap and plentiful, firms have little reason to invest in machines. The demographic dividend needs fertility to fall.

4 A worse future? Cities without factories

STRUCTURAL CHANGE

Africa is urbanizing without industrializing: its cities are fed by resource exports, not manufacturing. And automation (supported by AI) may be closing the factory route to economic development now.

Sources: World Bank population density, 2023 (via TheGlobalEconomy.com) and income groups; Acemoglu, Johnson and Robinson, QJE (2002); Gollin, Jedwab and Vollrath, J. Econ. Growth (2016); Rodrik, J. Econ. Growth (2016); The Economist review of How Africa Works (2026).

You be the prof



Read the article summary. As a group, write questions you think I might ask about this article and how it relates to the growth model. Make them a mix of difficulty levels. Draft the answers you want to hear.

I'll choose a group at random. The chosen group runs the discussion from the front: ask, call on classmates, push back, and close with a takeaway.

Ottawa wants a permanent off-switch for strikes

The Carney government plans to change Section 107 of the Canada Labour Code as part of its fall economic bill, the Building Canada Strong Act, according to government sources speaking to CBC News. Section 107 lets the federal government step into labour disputes in federally regulated industries such as railways, ports and airlines. The government can end a strike or lockout, send workers back to work, and have a neutral arbitrator impose the terms of the new contract instead of leaving the union and employer to bargain them. The section has existed since 1984 but was rarely used until recent years, when Ottawa used it to end a rail shutdown, stoppages at the Montreal and Vancouver ports, and the Air Canada flight attendants' strike in August 2025. The new bill would write this power explicitly into law, focused on disputes that could seriously disrupt exports. The government's stated goal is to reassure businesses and investors that Canada is a reliable place to invest. Unions argue it weakens workers' right to strike, which is their main source of bargaining power.



Source: Kate McKenna, CBC News, September 18, 2026

Ottawa wants a permanent off-switch for strikes

The Carney government plans to change Section 107 of the Canada Labour Code as part of its fall economic bill, the Building Canada Strong Act, according to government sources speaking to CBC News. Section 107 lets the federal government step into labour disputes in federally regulated industries such as railways, ports and airlines. The government can end a strike or lockout, send workers back to work, and have a neutral arbitrator impose the terms of the new contract instead of leaving the union and employer to bargain them. The section has existed since 1984 but was rarely used until recent years, when Ottawa used it to end a rail shutdown, stoppages at the Montreal and Vancouver ports, and the Air Canada flight attendants' strike in August 2025. The new bill would write this power explicitly into law, focused on disputes that could seriously disrupt exports. The government's stated goal is to reassure businesses and investors that Canada is a reliable place to invest. Unions argue it weakens workers' right to strike, which is their main source of bargaining power.



Source: Kate McKenna, CBC News, September 18, 2026

Buzz up! 6 min

Protection that backfires

In an editorial, the Wall Street Journal's editorial board argues that President Trump's new tariffs of up to 100% on patented medicines and their ingredients will hurt the U.S. biotech industry they are meant to protect. Most large drug makers have avoided the tariffs by cutting deals with the administration, promising to expand U.S. manufacturing and sell some drugs to Medicaid at low prices. Small biotech firms can't easily do the same. Most fund their research from sales of one or two drugs that treat small numbers of patients, and more than 90% rely on contract manufacturers, many of them abroad, because a small firm can't justify spending billions on its own plant. Moving production home is also slow: a new bio-manufacturing facility takes about seven years to build in the U.S., compared with two in China and three in Ireland. The administration has exempted some categories, such as rare-disease drugs and gene therapies. But firms must apply product by product and show an “urgent U.S. health need,” with no timeline for a decision, and tariff rates can shift as trade deals change. One small firm estimates compliance could consume 2–4% of its working capital before it sells its first drug. The editorial concludes that the added uncertainty will reduce U.S. biotech investment and R&D, raise drug prices, and help China, which has just announced a five-year plan to dominate the industry.



OPINION • REVIEW & OUTLOOK

The U.S. Tariff Attack on U.S. Biotech

New Trump border taxes arrive Tuesday and are a gift to Xi Jinping.

By The Editorial Board

Sept. 27, 2026 4:11pm EDT

President Donald Trump MATT KAMINSKY/ZUMA PRESS

Source: The Editorial Board, Wall Street Journal, September 27, 2026

Protection that backfires

In an editorial, the Wall Street Journal's editorial board argues that President Trump's new tariffs of up to 100% on patented medicines and their ingredients will hurt the U.S. biotech industry they are meant to protect. Most large drug makers have avoided the tariffs by cutting deals with the administration, promising to expand U.S. manufacturing and sell some drugs to Medicaid at low prices. Small biotech firms can't easily do the same. Most fund their research from sales of one or two drugs that treat small numbers of patients, and more than 90% rely on contract manufacturers, many of them abroad, because a small firm can't justify spending billions on its own plant. Moving production home is also slow: a new bio-manufacturing facility takes about seven years to build in the U.S., compared with two in China and three in Ireland. The administration has exempted some categories, such as rare-disease drugs and gene therapies. But firms must apply product by product and show an “urgent U.S. health need,” with no timeline for a decision, and tariff rates can shift as trade deals change. One small firm estimates compliance could consume 2–4% of its working capital before it sells its first drug. The editorial concludes that the added uncertainty will reduce U.S. biotech investment and R&D, raise drug prices, and help China, which has just announced a five-year plan to dominate the industry.



OPINION • REVIEW & OUTLOOK

The U.S. Tariff Attack on U.S. Biotech

New Trump border taxes arrive Tuesday and are a gift to Xi Jinping.

By The Editorial Board

Sept. 27, 2026 4:11pm EDT

President Donald Trump MATT KAMINSKY/ZUMA PRESS

Source: The Editorial Board, Wall Street Journal, September 27, 2026

Buzz up! 6 min

Why do the best firms leave home?

Mini case · FieldSense: Da Nang or Shenzhen?

LISTEN · A 1½-MINUTE PODCAST



FieldSense: should Linh leave Da Nang?

MINI CASE · DISCUSS IN GROUPS

1. Why would FieldSense be more productive in Shenzhen than in Da Nang?

2. Where do those advantages show up in our growth model?

3. What, if anything, should Vietnam's government do about it?